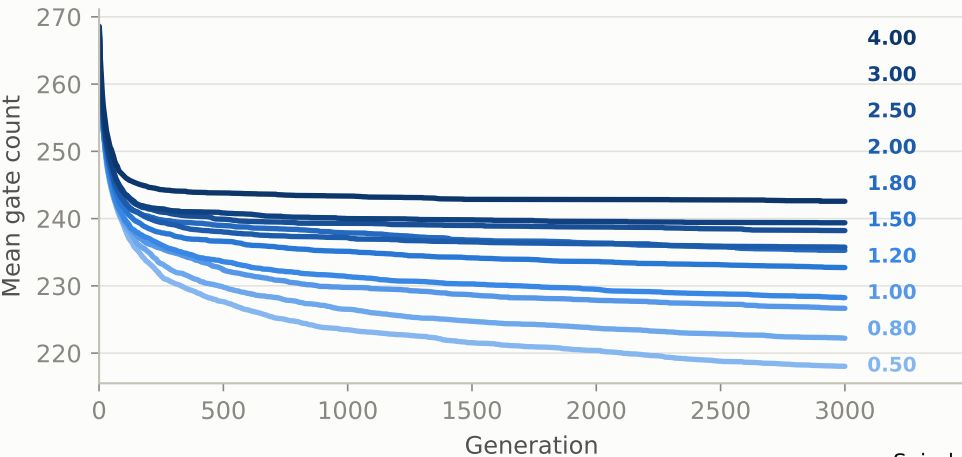
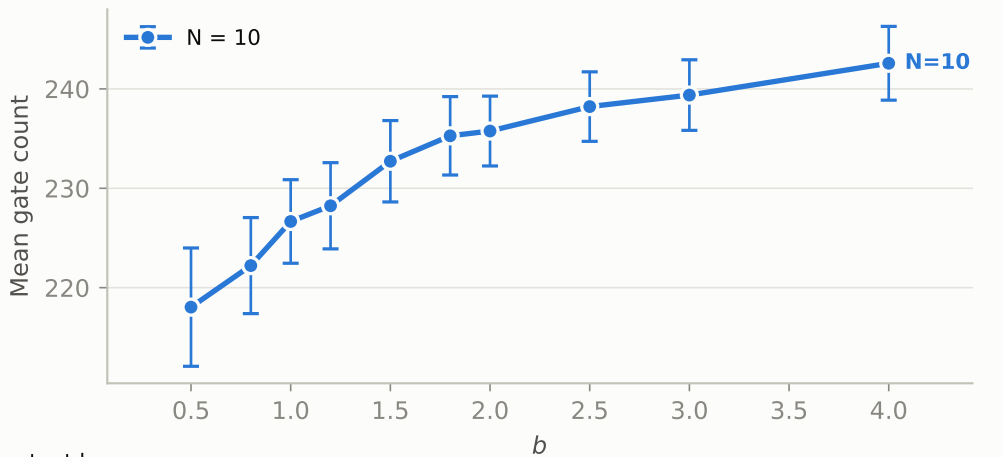


WOA parameter study — 6-bit problem #1, 100 trials × 3000 generations

(a) Population N = 10



(b) Final result vs spiral constant b (mean $\pm$ SD, gen 3000)



Spiral constant b

$b=0.50$	$b=1.00$	$b=1.50$	$b=2.00$	$b=3.00$
$b=0.80$	$b=1.20$	$b=1.80$	$b=2.50$	$b=4.00$